

EXPONENTIAL FUNCTIONS — GROWTH & DECAY

Digital SAT Style | Total: 60 Questions | Mixed Difficulty

● Easy: Q1–Q10 ● Medium: Q11–Q34 ● Hard: Q28–Q60

Section A — Recognising Exponential Models (Q1–Q10)

Q1. ● Easy

Which of the following functions represents exponential growth?

- A) $f(x) = 3x + 2$
- B) $f(x) = x^3 + 2$
- C) $f(x) = 3(1.2)^x$
- D) $f(x) = 3/(1.2x)$

Q2. ● Easy

A population of bacteria doubles every 3 hours. If the initial population is 500, which function models the population P after t hours?

- A) $P(t) = 500 \cdot 2^t$
- B) $P(t) = 500 \cdot 2^{(t/3)}$
- C) $P(t) = 500 + 2t$
- D) $P(t) = 500(3)^t$

Q3. ● Easy

The value of a car depreciates at a rate of 15% per year. Which function models the car's value V , in dollars, after t years, if the car was originally worth \$20,000?

- A) $V(t) = 20000 - 0.15t$
- B) $V(t) = 20000(0.85)^t$
- C) $V(t) = 20000(1.15)^t$
- D) $V(t) = 20000(0.15)^t$

Q4. ● Easy

The function $f(t) = 8(0.5)^t$ represents the amount, in grams, of a radioactive substance remaining after t years. What was the initial amount?

- A) 0.5 grams
- B) 4 grams
- C) 8 grams
- D) 16 grams

Q5. ● Easy

Which of the following represents exponential decay?

- A) $y = 5(1.3)^x$
- B) $y = 5(0.7)^x$

- C) $y = 5x^{0.7}$
D) $y = 5 - 0.7x$

Q6. Easy

The function $g(x) = 4(3)^x$ is evaluated at $x = 2$. What is $g(2)$?

- A) 24
B) 36
C) 48
D) 12

Q7. Easy

The function $h(t) = 200(0.5)^t$ models the amount of a substance in milligrams after t days. How many milligrams remain after 3 days?

GRID-IN (Enter your answer in the box provided)

Q8. Easy

An investment grows by 6% each year. If the initial investment is \$1,000, which expression gives the value after 5 years?

- A) $1000 + 0.06(5)$
B) $1000(1.06)^5$
C) $1000(0.06)^5$
D) $1000 \times 5 \times 0.06$

Q9. Easy

The graph of $f(x) = a \cdot b^x$ passes through $(0, 4)$. What is the value of a ?

- A) 0
B) 1
C) 4
D) b

Q10. Easy

Which of the following is the y-intercept of $f(x) = 7(2)^x$?

- A) 2
B) 7
C) 14
D) 0

Section B — Growth Rate & Decay Rate (Q11–Q22)

Q11. Medium

The function $P(t) = 1200(1.08)^t$ models a savings account balance in dollars after t years. By what percentage does the balance grow each year?

- A) 0.8%
- B) 8%
- C) 12%
- D) 80%

Q12. Medium

The function $N(t) = 5000(0.92)^t$ represents the number of fish in a lake after t years. At what rate is the fish population decreasing per year?

- A) 8%
- B) 9.2%
- C) 92%
- D) 0.92%

Q13. Medium

A town's population can be modeled by $P(t) = 12000(1.025)^t$, where t is in years. What is the approximate percent increase in population per year?

- A) 0.25%
- B) 2.5%
- C) 12.5%
- D) 25%

Q14. Medium

The function $V(t) = 16000(0.78)^t$ models the value of a machine, in dollars, after t years. What is the annual rate of depreciation as a percent?

GRID-IN (Enter your answer in the box provided)

Q15. Medium

If $f(x) = ab^x$ and $f(0) = 5$ and $f(1) = 15$, what is the value of b ?

- A) 1
- B) 3
- C) 5
- D) 15

Q16. Medium

The function $y = 3(b)^x$ is decreasing. Which of the following must be true?

- A) $b > 1$
- B) $b = 1$
- C) $0 < b < 1$
- D) $b < 0$

Q17. Medium

The population of a city is modeled by $P(t) = P_0(1 + r)^t$. If the population doubles in 10 years, which expression equals r ?

- A) $r = 2^{(1/10)} - 1$
- B) $r = 2^{10} - 1$
- C) $r = 10/2 - 1$
- D) $r = (1/2)^{10}$

Q18. Medium

A lab sample of 800 bacteria is treated with a substance that reduces the count by 20% every hour. How many bacteria remain after 3 hours?

- A) 409.6
- B) 512
- C) 320
- D) 256

Q19. Medium

The value of an antique doubles every 8 years. If it is worth \$600 today, what will it be worth in 24 years?

GRID-IN (Enter your answer in the box provided)

Q20. Medium

A function f satisfies $f(0) = 10$ and decreases by 30% for each unit increase in x . Which equation defines f ?

- A) $f(x) = 10(0.3)^x$
- B) $f(x) = 10(0.7)^x$
- C) $f(x) = 10(1.3)^x$
- D) $f(x) = 10 - 0.3x$

Q21. Medium

The half-life of a radioactive isotope is 5 years. If 80 grams are present initially, how many grams remain after 15 years?

- A) 5
- B) 10
- C) 20
- D) 40

Q22. Medium

A biologist records 250 cells at $t = 0$. The number of cells triples every 4 hours. Which function gives the number of cells after t hours?

- A) $N(t) = 250(3)^t$
- B) $N(t) = 250(3)^{(4t)}$
- C) $N(t) = 250(3)^{(t/4)}$
- D) $N(t) = 250(4)^{(t/3)}$

Section C — Reading & Interpreting Exponential Models (Q23–Q34)

Q23. Medium

The function $T(d) = 98(0.5)^{(d/6)}$ models the temperature, in degrees, of a cooling object d minutes after it is removed from heat. What does the 6 represent in this context?

- A) The initial temperature
- B) The rate at which temperature decreases per minute
- C) The number of minutes for the temperature to halve
- D) The final temperature after cooling

Q24. Medium

The graph of $y = a(b)^x$ passes through (1, 12) and (2, 36). What is the value of b ?

- A) 2
- B) 3
- C) 4
- D) 6

Q25. Medium

An exponential function f has $f(0) = 4$ and $f(2) = 36$. What is the value of $f(1)$?

GRID-IN (Enter your answer in the box provided)

Q26. Medium

If $f(x) = 5(2)^x$ and $g(x) = 5(2)^{(x+3)}$, what is the ratio $g(x)/f(x)$?

- A) 3
- B) 5
- C) 8
- D) 15

Q27. Medium

The function $P(t) = 2400(0.96)^{(4t)}$ models the number of particles in a sample after t hours. At what percent rate per hour does the number of particles decrease?

- A) 4%
- B) 4.96%
- C) 14.8%
- D) 16%

Q28. Hard

The function $N(t) = 600 \cdot 2^{(t/3)}$ gives the number of bacteria t hours after noon. At what time does the number of bacteria first reach 4,800?

- A) 3:00 PM
- B) 6:00 PM
- C) 9:00 PM

D) 12:00 AM (midnight)

Q29. ● Hard

Two functions are defined as $f(x) = 100(1.05)^x$ and $g(x) = 100(1.05)^{(x-2)}$. Which of the following is true?

- A) $g(x) = f(x) + 2$
- B) $g(x) = f(x) - 2$
- C) $g(x) = f(x)/1.05^2$
- D) $g(x) = f(x) \times 1.05^2$

Q30. ● Hard

The function $f(t) = A \cdot b^t$ satisfies $f(3) = 40$ and $f(6) = 320$. What is the value of b ?

- A) 2
- B) 4
- C) 8
- D) 40

Q31. ● Hard

The function $f(x) = a(3)^x$ passes through $(2, 72)$. What is the value of a ?

GRID-IN (Enter your answer in the box provided)

Q32. ● Hard

A quantity grows exponentially. It is 150 at time $t = 2$ and 600 at time $t = 4$. What was the quantity at $t = 0$?

- A) 37.5
- B) 50
- C) 75
- D) 100

Q33. ● Hard

The function $P(t) = 800e^{(0.04t)}$ models a population. Which of the following functions gives the same initial population but grows twice as fast?

- A) $P(t) = 800e^{(0.02t)}$
- B) $P(t) = 1600e^{(0.04t)}$
- C) $P(t) = 800e^{(0.08t)}$
- D) $P(t) = 800e^{(2t)}$

Q34. ● Hard

The function $V(x) = 1000(1.1)^x$ is equivalent to $V(x) = 1000(b)^{(x/12)}$ for monthly compounding. What is the value of b , rounded to the nearest hundredth?

- A) $b \approx 1.01$
- B) $b \approx 1.08$
- C) $b \approx 1.10$

D) $b \approx 1.92$

Section D — Comparing, Rewriting & Transforming (Q35–Q46)

Q35. Medium

The function $f(x) = 2^x$ and $g(x) = 2^x + 5$. How does the graph of g differ from the graph of f ?

- A) g is shifted 5 units to the right
- B) g is shifted 5 units to the left
- C) g is shifted 5 units upward
- D) g is stretched vertically by a factor of 5

Q36. Medium

The function $f(x) = 3^x$ and $g(x) = 3^{x-2}$. The y -intercept of g is:

- A) $\frac{1}{9}$
- B) $\frac{1}{3}$
- C) 1
- D) 9

Q37. Medium

Which transformation maps $f(x) = 4^x$ to $g(x) = 4^{-x}$?

- A) Vertical stretch by factor of 4
- B) Reflection over the y -axis
- C) Reflection over the x -axis
- D) Horizontal shift of 4 units

Q38. Medium

The function $f(x) = 5(2)^x$ is rewritten as $f(x) = 5 \cdot 8^{(x/k)}$. What is the value of k ?

- A) 1
- B) 2
- C) 3
- D) 8

Q39. Hard

Two savings accounts start with \$1,000. Account A earns 5% annual interest. Account B earns 2.5% interest every 6 months. Which account has more money after 1 year and by how much?

- A) Account A by \$0.625
- B) Account B by \$0.625
- C) They are equal
- D) Account B by \$1.25

Q40. **Hard**

The function $f(x) = 2 \cdot 9^x$ can be written as $f(x) = 2 \cdot 3^{(kx)}$. What is k ?

- A) 1
- B) 2
- C) 3
- D) 9

Q41. **Hard**

The graph of $f(x) = 3^x$ is shifted 2 units to the right and 4 units down. What is $f(2)$ for the transformed function?

GRID-IN (Enter your answer in the box provided)

Q42. **Hard**

Which of the following equations is equivalent to $y = 4^{(2x)}$?

- A) $y = 2^{(4x)}$
- B) $y = 16^x$
- C) $y = 8^x$
- D) $y = 2^x$

Q43. **Hard**

A colony of bacteria grows from 200 to 1,600 in 6 hours. A second colony grows from 50 at the same rate. How many bacteria are in the second colony after 6 hours?

- A) 100
- B) 200
- C) 400
- D) 600

Q44. **Hard**

The graph of $f(x) = 6(1/2)^x$ and $g(x) = 6(2)^x$ intersect at $x = 0$. For $x > 0$, which function is greater?

- A) $f(x) > g(x)$ for all $x > 0$
- B) $g(x) > f(x)$ for all $x > 0$
- C) They are equal for all $x > 0$
- D) It depends on the value of x

Q45. **Hard**

The function $f(x) = a \cdot b^x$ satisfies $f(2) = 18$ and $f(5) = 486$. What is the value of $a + b$?

- A) 4
- B) 5
- C) 6
- D) 9

Q46. **Hard**

If $2^x = 32$ and $3^y = 81$, what is the value of $x + y$?

Section E — Word Problems & Real-World Contexts (Q47–Q60)

Q47.  **Medium**

A car bought for \$25,000 loses 18% of its value each year. Which expression gives the value after 4 years?

- A) $25000(0.18)^4$
- B) $25000(0.82)^4$
- C) $25000(1.18)^4$
- D) $25000 - 4(0.18)(25000)$

Q48.  **Medium**

A savings account starts at \$3,000 and earns 4% annual interest compounded annually. Which function gives the account balance B after t years?

- A) $B(t) = 3000(0.04)^t$
- B) $B(t) = 3000 + 0.04t$
- C) $B(t) = 3000(1.04)^t$
- D) $B(t) = 3000(1 + 4t)$

Q49.  **Medium**

A town had a population of 10,000 in 2010. The population grows by 3% per year. What is the population in 2020, rounded to the nearest whole number?

GRID-IN (Enter your answer in the box provided)

Q50.  **Medium**

A scientist tracks radioactive decay with the model $A(t) = 240(0.5)^{(t/20)}$, where t is in years. After how many years will 30 grams remain?

- A) 20 years
- B) 40 years
- C) 60 years
- D) 80 years

Q51.  **Hard**

An investment of \$5,000 earns interest compounded annually. After 3 years the balance is \$5,955.08. What is the annual interest rate, to the nearest whole percent?

- A) 4%
- B) 5%
- C) 6%
- D) 7%

Q52.  **Hard**

A tablet computer is initially priced at \$800 and loses 25% of its value each year. After how many years is its value first below \$200?

- A) 4
- B) 5
- C) 6
- D) 7

Q53. **Hard**

The number of active users of an app is modeled by $U(t) = 4000(1.15)^t$, where t is the number of months after launch. During which month does the number of users first exceed 10,000?

- A) Month 5
- B) Month 6
- C) Month 7
- D) Month 8

Q54. **Hard**

A culture of yeast doubles every 90 minutes. If there are 500 cells at 12:00 PM, how many cells are there at 4:30 PM?

GRID-IN (Enter your answer in the box provided)

Q55. **Hard**

The function $P(t) = 8000(0.85)^t$ models the number of subscribers to a service. By approximately what year will the number of subscribers fall below 1,000, if $t = 0$ corresponds to 2024?

- A) 2037
- B) 2039
- C) 2041
- D) 2044

Q56. **Hard**

A colony starts with n_0 bacteria and grows at rate r per hour. After 2 hours there are 900 bacteria, and after 4 hours there are 8,100. What is n_0 ?

- A) 90
- B) 100
- C) 150
- D) 300

Q57. **Hard**

A radioactive substance has a half-life of 12 years. What fraction of the original amount remains after 36 years? Give your answer as a decimal.

GRID-IN (Enter your answer in the box provided)

Q58. **Hard**

The value of a painting increases by 8% per year. How many years, to the nearest whole year, will it take for the value to triple?

- A) 13
- B) 14
- C) 15
- D) 16

Q59. **Hard**

A function models compound interest: $A = P(1 + r/n)^{nt}$. For $P = 2000$, $r = 0.06$, $n = 12$, and $t = 5$, which expression best approximates A ?

- A) $2000(1.005)^{60}$
- B) $2000(1.06)^5$
- C) $2000(1.0605)^5$
- D) $2000(1 + 0.3)$

Q60. **Hard**

Two towns each have 5,000 residents. Town A grows by 4% per year; Town B shrinks by 2% per year. After approximately how many years will Town A's population first be more than double Town B's population?

- A) 12
- B) 14
- C) 17
- D) 20